

CUSTOMER MILESTONE BRIEF

MS #36 - COBOL Qualification Hardening

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.36.0
Date	2026-08-30
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #36 - COBOL Qualification Hardening converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Defines what the platform can and cannot claim across the discovered COBOL estate and separates static qualification from native execution.

Why this matters to a customer

Defines what the platform can and cannot claim across the discovered COBOL estate and separates static qualification from native execution.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a graph-bound COBOL qualification contract covering 55 programs, 927 paragraphs, 75 copybooks, 7,817 fields, 56 file handles, and their static dependency relationships.
- Separated nine qualification gates for inventory, syntax/source format, copybook closure, data layout/numerics, control flow/calls, file behavior, DB2/CICS/IMS boundaries, local differential behavior, and authorized native compile/link/execute evidence.
- Added a COBOL compatibility ledger covering source format, copybooks, PIC layouts, REDEFINES/OCCURS, decimal behavior, arithmetic, control flow, linkage, file status, embedded DB2, CICS, IMS, compiler options, and LE runtime semantics.
- Required every COBOL boundary to resolve to exactly one of the five governing classifications and rejected silent policy acceptance or unsupported behavior without explicit exclusion.
- Kept native compiler qualification, runtime equivalence, mainframe equivalence, and production readiness false; generated COBOL planning cells remain planning scopes only.

How it differs from earlier work

MS #36 builds on MS #35 (DB2 Semantic Adapter Hardening). The earlier milestone established its own bounded capability; this milestone adds cobol qualification hardening while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use cobol qualification hardening as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- Kept native compiler qualification, runtime equivalence, mainframe equivalence, and production readiness false; generated COBOL planning cells remain planning scopes only.

Source of record

- CHANGELOG.md release section(s): 0.36.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity